Huawei Access Node JAY-ACC-68 Routing Protocol Flap Resolution SOP

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9334

Description

The Huawei Access Node JAY-ACC-68 Routing Protocol Flap failure domain is a critical event that occurs when the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding causes instability in the routing protocol, leading to a major alarm (ALARM_9334). This SOP is designed to troubleshoot and resolve this issue on the Huawei Access Node JAY-ACC-68 at the Jayanagar Metro Site. The root cause of this failure domain is attributed to the complex interactions between the routing protocols, including BGP, OSPF, and ISIS, which are used for routing and forwarding decisions. The Huawei Access Node JAY-ACC-68 is responsible for establishing and maintaining these routing protocols, and any disruptions or misconfigurations can cause a routing protocol flap, leading to packet loss, increased latency, and poor network performance. The SOP will guide the network engineer through a series of diagnostic commands, resolution steps, and verification commands to identify and fix the root cause of the routing protocol flap.

Pre-requisite Checks

- Check: Verify the site location and ensure remote hands are dispatched to the Jayanagar Metro Site., Description: Ensure that the network engineer has access to the Huawei Access Node JAY-ACC-68 and all necessary equipment to perform the troubleshooting and resolution steps.

- Check: Verify the redundant link status and ensure that the link is stable and functioning correctly., Description: Verify that the redundant link is up and operational to ensure that the network is not impacted during the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP process is running, and the BGP neighbor relationship is established., Description: This command displays information about the BGP process, including the BGP version, AS number, and neighbor relationships. The expected output indicates that the BGP process is running and that the BGP neighbor relationship is established.
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```
Command: display ospf process, Expected Output: OSPF process is
running, and the OSPF neighbor relationship is established.,
Description: This command displays information about the OSPF process,
including the OSPF version, AS number, and neighbor relationships. The
expected output indicates that the OSPF process is running and that
the OSPF neighbor relationship is established.

Command: display isis process, Expected Output: ISIS process is
running, and the ISIS neighbor relationship is established.,
Description: This command displays information about the ISIS process,
including the ISIS version, AS number, and neighbor relationships. The
expected output indicates that the ISIS process is running and that
the ISIS neighbor relationship is established.
```

Resolution Steps

Step 1: Step: Verify the BGP configuration, Command: display bgp configuration, Description: Verify that the BGP configuration is correct and that there are no misconfigurations that could be causing the routing protocol flap.

Step 2: Step: Verify the OSPF configuration, Command: display ospf configuration, Description: Verify that the OSPF configuration is correct and that there are no misconfigurations that could be causing the routing protocol flap.

Step 3: Step: Verify the ISIS configuration, Command: display isis configuration, Description: Verify that the ISIS configuration is correct and that there are no misconfigurations that could be causing the routing protocol flap.

Step 4: Step: Check for routing protocol flapping, Command: display routing protocol flapping, Description: Check for any routing protocol flapping that could be causing the issue.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP process is running,
and the BGP neighbor relationship is established., Description: Verify
that the BGP process is running and that the BGP neighbor relationship
is established.

Command: display ospf process, Expected Output: OSPF process is
running, and the OSPF neighbor relationship is established.,
Description: Verify that the OSPF process is running and that the OSPF
neighbor relationship is established.

Command: display isis process, Expected Output: ISIS process is
running, and the ISIS neighbor relationship is established.,
Description: Verify that the ISIS process is running and that the ISIS
neighbor relationship is established.
```

Roll-back Procedures

- Step: Restore the original BGP configuration, Command: system-view > bgp 64512, Description: Restore the original BGP configuration to its previous state.

- Step: Restore the original OSPF configuration, Command: system-view > ospf 100, Description: Restore the original OSPF configuration to its previous state.

- Step: Restore the original ISIS configuration, Command: system-view > isis level-2, Description: Restore the original ISIS configuration to its previous state.